Bhukya Neeraj Nayak

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.09
SRI CHAITANYA JUNIOR COLLEGE	Class XII (Board of Intermediate Education Andhra Pradesh)	2023	83.80%
BHASHYAM (EM) HIGH SCHOOL	Class X (Board of Secondary Education Andhra Pradesh)	2021	99.33%

SCHOLASTIC ACHIEVEMENTS

Mechanical Design Internship:

- Completed a Mechanical Engineering internship at Vijay Innovation Technology, Bengaluru, focused on CAD-based product design and engineering drawings using Autodesk Fusion 360.

POSITIONS OF RESPONSIBILITY

Milan Events Coordinator, IIT Hyderabad :

- Coordinated and managed event activities for **Milan, IIT Hyderabad**, ensuring smooth planning, execution, and coordination across multiple teams.
- Managed on-ground event operations, logistics, scheduling, and coordination with student teams and stakeholders.
- Supported the successful execution of institute-level events while handling multiple responsibilities under tight timelines.

SKILLS

Modeling & Analysis Softwares: ANSYS (Fluent, Workbench), Solid Edge,AutoCAD,MATLAB

Soft Skills : Communication,Leadership,Decision-making,Team player,Management

RELEVANT COURSES

Design & Solid Mechanics: Finite Element Analysis (FEA), DfMA, Solid Mechanics, Machine Drawing

Thermo-Fluids & Simulation: Computational Fluid Dynamics (CFD), Thermodynamics, Fluid Mechanics, Heat Transfer

Systems & Manufacturing: Kinematics & Dynamics, Manufacturing Science, Control Systems

PROJECTS

3D Printed Scale Ship Model | CAD & Additive Manufacturing

- Designed a complex vessel shell CAD model and fabricated a high-precision physical prototype using polymer 3D printing.
- Created the ship model with detailed geometric features in CAD.
- Prepared the model for 3D printing and produced the physical prototype.

RC Hovercraft with Ducted Fan Drive | Propulsion & Design

- Modeled and assembled an RC hovercraft with a ducted fan thrust system, optimizing weight distribution for air-cushion lift.
- Designed the body and propulsion arrangement for the hovercraft.
- Assembled and tested the model for stable air-cushion movement.

2D Turbulent Flame Combustion Simulation | CFD

- Simulated a turbulent hydrogen-air flame in ANSYS Fluent using the Eddy Dissipation Model to analyze thermal and species contours.
- Set up the combustion simulation model in ANSYS Fluent.
- Studied temperature and species distribution using simulation contours.

Conveyor Belt with Geneva Drive | Design & Fabrication

- Modeled and fabricated a working mechanical prototype of an intermittent conveyor system driven by a Geneva mechanism.
- Designed the conveyor components and Geneva mechanism for intermittent motion.
- Assembled and tested the prototype to check its working motion.

EXTRACURRICULAR

Sports & Athletics: Active participant in campus sports, playing regular matches of Cricket, Football, and Badminton in inter-hostel tournaments.

Competitive & Casual Gaming : Interested in strategy, simulation, and tactical video games, with participation in local and online casual gaming sessions.

NSS Volunteer:

- Actively participated as an NSS volunteer at IIT Hyderabad, contributing to community service, social awareness, and various institute outreach activities.